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Fast ionic conductivity by thermal treatment with ultralow electronic transport in solid-state electrolyte Na_3YCl_6

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ABSTRACT

Improving the ionic and electronic conductivities of solid-state electrolytes (SSEs) is urgently needed to develop commercially viable all-solid-state batteries. Here, we provide atomistic insights into the electronic transport properties and Na ionic conductivity of the halide-based SSE Na_3YCl_6 (NYC) with a trigonal structure and propose a way for improving ionic conductivity by amorphization. Our *ab initio* calculations, employing a highly accurate hybrid functional and many-body method, reveal high electric and thermal insulating behavior of crystalline NYC. Using machine learning interatomic potential-based molecular dynamics simulations, we demonstrate low ionic conductivity at room temperature in the crystalline phase, but a significantly higher value of 0.29 mS/cm in amorphous NYC simulated by thermal treatment, highlighting that amorphization is an effective way for improving ionic conductivity.

With increasing demands for clean energy and electric vehicles, the development of high-performance, low-cost batteries has been rapidly promoted over the past decade.[1–3] Although the lithium-ion batteries (LIBs) dominate the battery market at present, their sustainable development is hindered by resource depletion, environmental damage, and serious safety issues.[4–9] As a potential candidate for LIBs, therefore, sodium-ion batteries (SIBs) have been drawing significant attention owing to an abundance of Na resources.[10,11] In particular, all-solid-state SIBs utilizing various solid-state electrolytes (SSEs) are expected to greatly increase energy density, realize much faster charging time, and ensure complete safety.[12–15] However, the key factors for commercially viable SSEs, such as fast ionic conductivity, wide electrochemical stability, and low electronic conductivity, are still unsatisfactory, remaining a major challenge for battery developers.[16,17]

Starting with the ground-breaking work of Asano *et al.*[18] with Li_3YX_6 (X = Cl, Br), halide-based SSEs have stimulated special interest due to their fast ionic conductivities and fairly broad electrochemical stability windows.[19,20] Compared with the multivalent sulfur or oxygen anions, the monovalent halogen anions with larger ionic radii induce weaker Coulombic interaction and longer ionic bond length with alkali cations, thereby leading to the fast ionic conductivities over $\sim$1 mS/cm.[21,22] Although various lithium halide SSEs exhibit very high ionic conductivities, their sodium analogs exhibit relatively low Na ionic conductivities (σ_{Na}).[23–27] For the cases of Na_3YCl_6 (NYC) and Na_2ZrCl_6, the room temperature (RT) σ_{Na} values were measured to be 9.5×10^{-5} and 1.4×10^{-4} mS/cm, respectively.[28–30] Using *ab initio* molecular dynamics (AIMD) simulations, Qie *et al.*[31] predicted relatively high RT σ_{Na} values of 0.77 and 0.44 mS/cm for NYC and Na_3YBr_6, but suggested a different crystal structure with a *P*321 space group from the experimental ones with an $R\bar{3}$[32] or $P2_1/n$ space group.[28] The low ionic conductivities in the Na-halides may be attributed to the stronger Na–X interaction and narrower migration channels compared with those in the Li-halides.

To improve the ionic conductivity in Na-halides, disordering or amorphization has been proposed as a promising strategy.[33–38] Wu *et al.*[28] measured enhanced σ_{Na} of 6.6×10^{-2} mS/cm in $Na_{2.25}Y_{0.25}Zr_{0.75}Cl_6$ (NYZC0.75), attributing it to the local structural disorder and/or non-stoichiometry on the cation lattice. Notably, Hu *et al.*[35] achieved exceptionally high σ_{Na} of 4 mS/cm at RT, even higher than those of Li-halides, in amorphous $NaTaCl_6$ (NTC), arising from a unique amorphous $TaCl_6$ octahedral network and weakening Na–Cl interactions. Furthermore, amorphous NTC showed remarkable mechanical deformability and electrochemical stability.[35] In addition, amorphization causes significant reduction of electronic conductivity (σ_e), being beneficial to the prevention of self-discharge in electrodes.[39,40] In contrast with that the high σ_e of SSE promotes the growth of dendrites in electrode;[41] disordered NYZC0.75 and amorphous NTC were found to be electronic insulators ($\sigma_e \approx 10^{-9}$ S/cm).[28,35] In this context, accurate quantitative evaluation of electronic conductivity and mechanical properties of SSEs is of importance in developing all-solid-state SIBs.

In this work, we report an ultralow electronic conductivity of crystalline Na_3YCl_6 (c-NYC) in a trigonal lattice with a space group $R\bar{3}$ and fast Na ionic conductivity of its amorphous phase (a-NYC) by applying the state-of-the-art *ab initio* methodology. We calculated the accurate electronic band structure by applying the HSE06 method[42] using the VASP code[43] and the GW method using the BerkeleyGW code[44] in connection with the Quantum ESPRESSO (QE) package.[45] The electronic transport properties were obtained from the electron scattering rates calculated by considering acoustic deformation potential (ADP), polar optical phonon (POP), and ionized impurity (IMP) as implemented in the AMSET code[46] in conjunction with VASP. Using the training data obtained from the AIMD simulations performed at various temperatures, we constructed a highly accurate machine-learning interatomic potential (MLIP)[47] for classical MD (CMD) simulations as implemented in the LAMMPS code[48] to evaluate the Na ionic conductivity in the amorphous phase. Computational details are provided in the supplementary material.

First, we identified the crystal structure of c-NYC, which was known to be crystallized in the trigonal system with an $R\bar{3}$ space group (mp-675104[49]) by experiment.[32] Since Na atoms have partial occupancies of 0.513 (Na1) and 0.487 (Na2), we generated two distinctive structures and performed structural optimizations to identify the real structure with lower energy. It turned out that the structure with a full occupancy at Na1 site is the ground-state shown in Fig. 1(a) (see Table S1 in the supplementary material). In this structure, both Y and Na atoms are sixfold coordinated with Cl atoms, forming edge-sharing $NaCl_6$–$NaCl_6$ and $NaCl_6$–YCl_6 octahedral network on the *xy* plane and plane-sharing $NaCl_6$–$NaCl_6$ octahedral zig-zag chain in the *z* direction. Moreover, there are two empty cation (Na2) vacancy sites around Na atoms at the Na3 sites, which can act as intermediate sites for Na migration, giving an expectation of fast ionic conductivity.

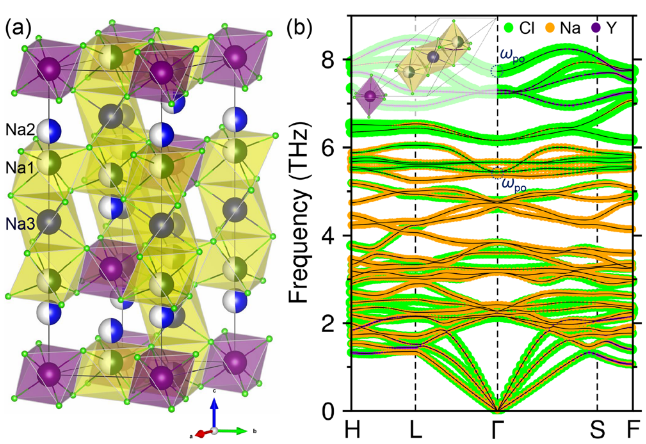

FIG. 1. (a) Polyhedral view of the conventional unit cell of Na_3YCl_6 (space group $R\bar{3}$), where Na has partial occupancies of 0.513 (Na1) and 0.487 (Na2), and Y and Cl are denoted by violet- and green-colored balls, respectively. (b) Phonon dispersion curve with atomic contributions, where the inset shows the primitive unit cell and dotted circles denote the major modes for effective phonon frequency ω_{po}.

We transformed the conventional unit cell (3 formula units, 30 atoms) in the hexagonal representation into the primitive unit cell (1 fu, 10 atoms) in the rhombohedral representation [see Fig. 1(b) inset]. The optimized lattice parameters of the primitive unit cell are $a = 7.459$ Å and $\alpha = 56.132°$, in good agreement with the experimental values of $a = 7.416$ Å and $\alpha = 56.087°$[32] within the small relative errors of 0.57% and 0.08%, respectively (see Table S1 in the supplementary material). To confirm the dynamical and mechanical stabilities, we calculated phonon dispersion and elastic constants, which are used in further calculations of electron scattering rate. As shown in Fig. 1(b), no soft modes with imaginary phonon frequencies were observed in the calculated phonon dispersion curve, indicating that c-NYC is dynamically stable in the trigonal system (see Fig. S1 for the conventional unit cell in the supplementary material). Furthermore, the calculated elastic constants were found to meet the mechanical stability criteria[50] (see Table S2 in the supplementary material).

From the calculated elastic constants, we evaluated the elastic moduli of polycrystalline solids within the different approximations (see Table S2 in the supplementary material). We first highlight that the shear moduli G of c-NYC were determined to be 8.99–9.17 GPa, nearly twice as high as the shear moduli of Na metal (0.63–4.34 GPa) in various crystalline phases listed in the Materials Project (MP) database.[49] Considering the suggestion that the shear modulus of electrolyte should be at least twice larger than that of Li metal to prevent the growth of dendrites by linear elasticity analysis in LIBs,[51] this indicates that c-NYC can effectively suppress the dendrite formation on the electrode. We are therefore interested in the Poisson's ratio ν and Pugh's ratio μ to check whether the material is ductile or fragile, since adequate ductility of SSE is required for mechanical compatibility with electrodes. For the c-NYC, the ν and μ values were determined to be 0.32 and 2.40, respectively, which are clearly larger than the threshold values of 0.26 and 1.75 for ductility,[52] indicating that c-NYC is surely ductile. All these mechanical features indicate the suitability of c-NYC as a promising ion conductor.

Then, we calculated the electronic band structure using the HSE06 hybrid functional method and the many-body GW method for bandgap E_g with high accuracy. Figure 2(a) shows the calculated electronic band structure and atom-projected partial density of states (DOS) (see Fig. S2 for DFT + U calculation in the supplementary material). The indirect bandgap E_g ($\Gamma \rightarrow$ F) was determined to be 6.938/7.176 eV using the HSE06/GW method, while it was 5.342 eV using the DFT + U method, which are wider than, while similar to, the previous calculation of 5.155 eV from the MP database.[49] It should be emphasized that the HSE06 hybrid functional yielded E_g value close to the GW method, indicating a high accuracy of the calculated bandgap. From the DOS analysis, the electronic states at the valence band maximum (VBM) and conduction band minimum (CBM) levels were found to be dominated by Cl 3p and Y 4d electrons, respectively. Moreover, the valence and conduction bands around VBM and CBM look quite flat, which indicates, together with the wide bandgap, that c-NYC is a highly insulating crystalline solid.

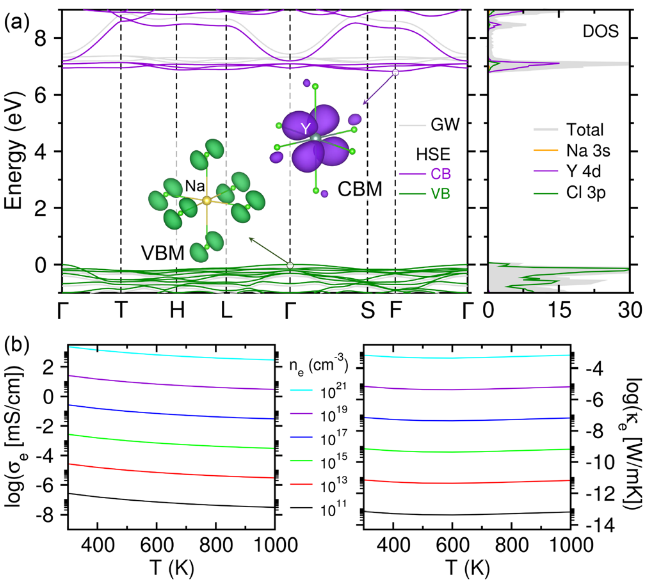

FIG. 2. (a) Electronic band structure and density of states (DOS in states/eV) of c-NYC calculated using HSE06 and GW methods (gray), with squared wave functions corresponding to the valence band maximum (VBM) and conduction band minimum (CBM). (b) Electron contribution to electrical (σ_e) and thermal (κ_e) conductivities as functions of temperature at different carrier concentrations n_e.

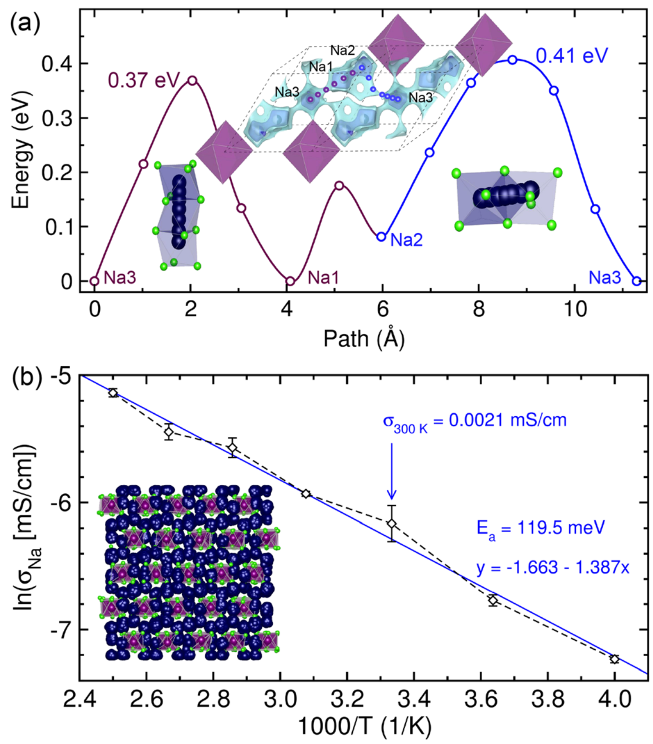

FIG. 3. (a) Energy profile for Na migrations along the straight pathway Na3–Na1–Na2 (violet color) and zigzag pathway Na2–Na3 (blue color) shown in the inset, calculated using the DFT-NEB method for c-NYC. (b) Arrhenius plot of Na ionic conductivity (σ_{Na}) determined from MLIP-CMD simulations for c-NYC, where the inset shows the equilibrium structure of the simulation box of a $6 \times 6 \times 2$ supercell.

Using the HSE06-calculated wave functions and outputs from the phonon, elastic, dielectric, and deformation potential calculations (see Table S3 in the supplementary material), we calculated the electronic transport properties as functions of temperature at different carrier concentrations. For the POP scattering, the effective phonon frequency ω_{po} was calculated to be 4.95 THz, with major contributions of the middle- and high-ranged (5.25 and 7.89 THz) optical phonon modes originated from Na and Cl atoms [see Fig. 1(b) and Table S4 in the supplementary material]. Through the analysis of mode-dependent electron scattering lifetime τ_{nk}, it was revealed that the scattering is dominated by POP modes with $\tau_{nk}^{-1} \approx 10^4\ \mathrm{ps}^{-1}$ both in low and high carrier concentrations (see Figs. S3 and S4 in the supplementary material). Moreover, the IMP mode was found to induce the lowest impact ($\tau_{nk}^{-1} \approx 0.01\ \mathrm{ps}^{-1}$) in low carrier concentration ($n_{e/h} = 10^{11}\ \mathrm{cm}^{-3}$), whereas the ADP mode had the lowest impact ($\tau_{nk}^{-1} \approx 20\ \mathrm{ps}^{-1}$) in high concentration ($n_{e/h} = 10^{20}\ \mathrm{cm}^{-3}$). Therefore, the charge carrier e/h mobilities are almost determined by the POP mode (see Figs. S5 and S6 in the supplementary material). Figure 2(b) shows the electron contributions to electric conductivity (σ_e) and thermal conductivity (κ_e) as functions of temperatures at different carrier concentrations (see Fig. S7 for the hole contributions in the supplementary material). At the carrier concentration of $n_e = 10^{13}\ \mathrm{cm}^{-3}$, the σ_e values changed from 2.70×10^{-8} to 3.03×10^{-9} S/cm according to the temperature from 300 to 1000 K, in reasonable agreement with the experimental value of 8.89×10^{-9} S/cm for NYZC0.75.[28] As the concentration decreases, the electronic conductivity was found to exponentially decrease, reaching the minimum value of 3.06×10^{-12} S/cm at $n_e = 10^8\ \mathrm{cm}^{-3}$, indicating the high electric insulating behavior of c-NYC. The thermal conductivity κ_e was also very low, suggesting that c-NYC is thermally insulating as well.

We then evaluated the Na ionic conductivity σ_{Na} to elucidate the most important property of SSE. To this end, Na ion migration channels were identified by plotting the bond valence site energy isosurface in c-NYC, as depicted in Fig. 3(a). Two pathways were identified in the rhombohedral primitive unit cell: (1) the pathway composed of the straight line along the [111] direction (Na3–Na1–Na2) and the crooked line (Na2–Na3) and (2) the zigzag-type pathway on the (111) plane (Na2–Na3–Na2). It is worth noting that for the first pathway, the YCl_6 octahedra block the straight motion of Na ion and thus the pathway is bent, resulting in step-like pathway. Since the second pathway is involved in the first one, we estimated the activation energy E_a along the first pathway. The E_a values were determined to be 0.37 and 0.41 eV with the lengths of 6.12 and 5.18 Å along the Na3–Na1–Na2 and Na2–Na3 segments, respectively. When compared with the previous calculations,[14] these values of E_a correspond to not so fast σ_{Na} of $\sim 10^{-3}$ mS/cm. To improve σ_{Na}, therefore, we suggest following the strategy of disordering and/or amorphization by applying the large-scale CMD simulations.

To construct MLIP for CMD simulations, we built the $2 \times 2 \times 2$ supercells (120 atoms) of hexagonal unit cell (30 atoms) and performed AIMD simulations at elevating temperature from 300 to 1300 K with an interval of 200 K (see Table S5 and Fig. S8 for convergence). Using the constructed MLIP, we then performed the large-scale CMD simulations on the $6 \times 6 \times 2$ cell (2160 atoms) to evaluate

σ_{Na}. From the long-term (2 ns) *NVE* runs after *NPT* equilibrations at 250–400 K with a step of 25 K, we determined the self-diffusion coefficients of Na atoms (D_{Na}), from which σ_{Na} was derived using the Nernst–Einstein formula, while the activation energy E_a was determined using the Arrhenius equation. Figure 3(b) displays the Arrhenius plot of the ionic conductivities vs 1000/T at 250–400 K for c-NYC. The RT σ_{Na} value was determined to be 2.1×10^{-3} mS/cm ($E_a = 0.12$ eV), which is higher than the experimental value of 1.4×10^{-4} mS/cm in the monoclinic phase with $P2_1/n$ space group[28] but much lower than the theoretical value 0.77 mS/cm in the trigonal phase[31] and the critical value 1 mS/cm for a reasonable SSE. In the following, therefore, we inevitably considered a-NYC, which is expected to improve ionic conductivity as mentioned above.

To get disordered or amorphous phases of NYC, we performed the thermal treatment process on the simulation box. Before this process, we preliminarily determined the phase transition temperature T_{pt} by heating the box to 2000 K while measuring the volume. As the temperature increased, the box volume was gradually increased for a while, and a sudden volume change was observed at around 1000 K, indicating that the crystalline phase was transformed into the amorphous one (see Fig. S9 in the supplementary material). Based on this observation, we set the maximum heating temperature T_{max} as 600–1300 K with an interval of 100 K. Then, we performed the thermal treatment process; the box was first heated by increasing temperature from 0 to T_{max} at a heating rate of 1 K/ps, equilibrated for 3 ns at T_{max}, then cooled rapidly to RT for 100 ps, and finally equilibrated for 3 ns at RT (see Figs. S10 and S11 in the supplementary material). The *NPT* ensemble method was utilized at each step. We repeated the thermal treatment procedure three times with different random seeds for each T_{max}, averaging over them.

After the thermal treatment, we performed further *NVT* equilibrations for 3 ns and determined σ_{Na} values from the obtained trajectories. Figure 4(a) shows the RT Na ionic conductivities $\sigma_{Na,300\,K}$ of the heat-treated NYC samples as a function of T_{max}. When $T_{max} = 600$ K, the $\sigma_{Na,300\,K}$ value was 4.1×10^{-3} mS/cm, being close to the value of non-heated c-NYC. As T_{max} increased, the $\sigma_{Na,300\,K}$ was found to rapidly increase, reaching the maximum value of 0.29 mS/cm at $T_{max} = 1000$ K, and then slightly decreased to 0.19 mS/cm at $T_{max} = 1300$ K. Interestingly, significantly high $\sigma_{Na,300\,K}$ values were obtained when T_{max} was over the phase transition temperature $T_{pt} = 1000$ K of NYC, confirming that amorphization can remarkably improve the ionic conductivity of SSEs. In this work, the cooling rate increased from 3 to 10 K/ps with an interval of 1 K/ps as T_{max} increased from 600 to 1300 K, and the gradual decrease in $\sigma_{Na,300\,K}$ over $T_{max} = 1000$ K might be associated with such increase in the cooling rate.

To gain an atomistic insight into the fast Na ion conductivity, we performed structural analysis of the heat-treated structures obtained by thermal treatment. It was observed that the phases treated at $T_{max} < T_{pt}$ maintained the initial framework with clear confusion, thereby called disordered structures, but those treated at $T_{max} \geq T_{pt}$ showed fully destroyed framework, that is, amorphous phase (see Fig. S12 in the supplementary material). Intuitively, the disordered structures consist of mostly $[YCl_6]^{3-}$ octahedra, whereas the amorphous phases are characterized by various kinds of Y-centered clusters such as $[YCl_5]^{2-}$ and $[YCl_4]^{-}$ with their complex interconnections. To quantitatively estimate the disordering degree, we counted the remaining number of YCl_6 octahedron N_{YCl_6}, as shown in Fig. 4(b) (see Table S6 in the supplementary material). Compared with the crystalline phase ($N_{YCl_6} = 216$), the disordered structures showed only slight reduction ($N_{YCl_6} = 180 - 175$), but the amorphization induced sudden and remarkable reduction ($N_{YCl_6} = 56 - 77$). Considering the blocking role of YCl_6 as mentioned earlier, such destruction and reduction of YCl_6 octahedral network could allow the straight migration of Na ion rather than the crooked migration, resulting in the fast Na ion conduction in a-NYC. We analyzed the radial distribution function between various ion pairs for a more direct structural characterization (see Fig. S13 in the supplementary material). Among the peaks, the Y–Cl pairs exhibit the highest and sharpest peaks within 3 Å position in all the structures, indicating that they are well-ordered and difficult to move around due to the strong interaction between the Y^{3+} cation and Cl^- anions. The second sharpest peaks were found for the Na–Cl pairs with almost similar patterns in all the structures, implying that these pairs have moderate mobility. Most importantly, the Na–Na pairs show the broadest and lowest peaks, indicating their disordered distribution and high mobility. The Na–Na peaks in the disordered phases (T_{max} below 1000 K) were similar to that in the crystalline phase, whereas in the amorphous phases (T_{max} over 1000 K), they lacked periodicity, resembling that of a liquid, explaining the reason for higher Na ionic conductivity.

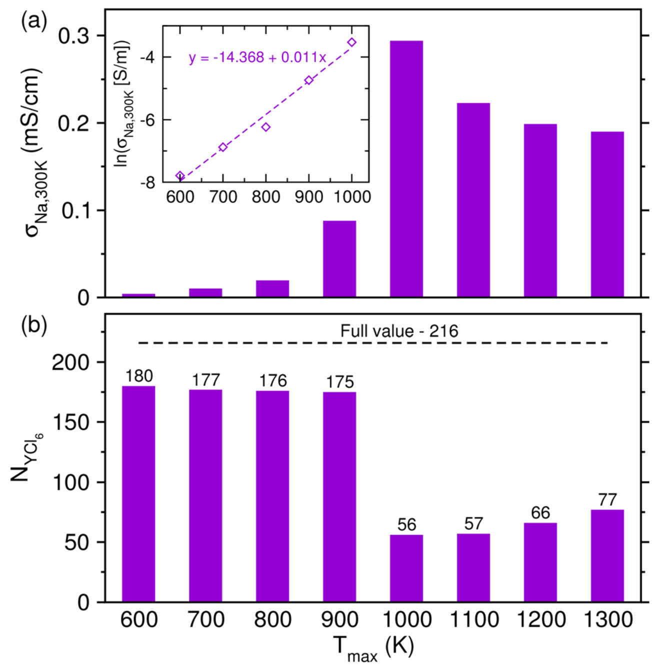

FIG. 4. (a) Na ionic conductivities at 300 K ($\sigma_{Na,300\,K}$) in disordered and amorphous NYC as varying T_{max} from 600 to 1300 K, where the inset shows the exponential relation of σ_{Na} vs T_{max} from 600 to 1000 K, and (b) the corresponding number of YCl_6 octahedra (N_{YCl_6}), where the horizontal dashed line denotes the full number in c-NYC.

In conclusion, we have investigated the structural, electronic, and ionic transport properties of crystalline and amorphous Na_3YCl_6 using highly accurate *ab initio* calculations. Confirming the thermodynamic and mechanical stabilities of the crystalline phase, our calculations revealed very low charge carrier (electron/hole) contributions to the electrical and thermal conductivities by considering the deformation, phonon, and impurity effects. We demonstrated the relatively low RT Na ionic conductivity of 2.1×10^{-3} mS/cm in the crystalline phase, in contrast to a significantly high value of 0.29 mS/cm in the amorphous phase simulated by thermal treatment with a maximum heating temperature of 1000 K, ascribable to the reduction in number of YCl_6 octahedra. This work provides a valuable atomistic insight into the ultralow electronic and fast ionic conductivities of halide-based SSEs, highlighting the feasibility of enhancing ionic conductivity by amorphization and developing all-solid-state SIBs.

See the supplementary material for computational details; tables for structural, elastic, optical phonon, and MLIP data; and figures for phonon/electronic band structures, transport properties, and CMD data.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Tae-Il Ri: Investigation (equal); Writing – original draft (equal). **Suk-Gyong Hwang:** Software (equal). **Jin-Song Kim:** Methodology (equal); Visualization (equal). **Kum-Chol Ri:** Data curation (equal); Methodology (equal). **Chol-Jun Yu:** Investigation (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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